

SHETH L.U.J AND SIR MV COLLEGE

PRACTICAL NO:-09

AIM: Hypothesis Testing:

- Formulate null and alternative hypotheses related to a business problem, conduct a hypothesis test using appropriate statistical tests, and interpret the results.
- Perform A/B testing on a website or marketing campaign to evaluate the effectiveness of different strategies and make data-driven decisions

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T128_P09.py - C:/Users/Rani/Downloads/T128_P09.py (3.13.7)
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import pandas as pd
import scipy.stats as stats
import matplotlib.pyplot as plt

# Student Details
print("Name: Rani Chavre")
print("Roll No.: T128")
print("AIM: Hypothesis Testing")

# Load Dataset
df = pd.read_csv("Rani_Hypothesis_Testing_AB_Dataset.csv")

print("\n--- DATASET ---")
print(df.head())

# Separate A/B groups
A = df[df["Marketing_Strategy"] == "Strategy A"]["Profit"]
B = df[df["Marketing_Strategy"] == "Strategy B"]["Profit"]

# Hypotheses
print("\n--- HYPOTHESIS ---")
print("H0: There is no significant difference between Strategy A and Strategy B.")
print("H1: There is a significant difference between Strategy A and Strategy B.")

# Average Profit
print("\n--- AVERAGE PROFIT ---")

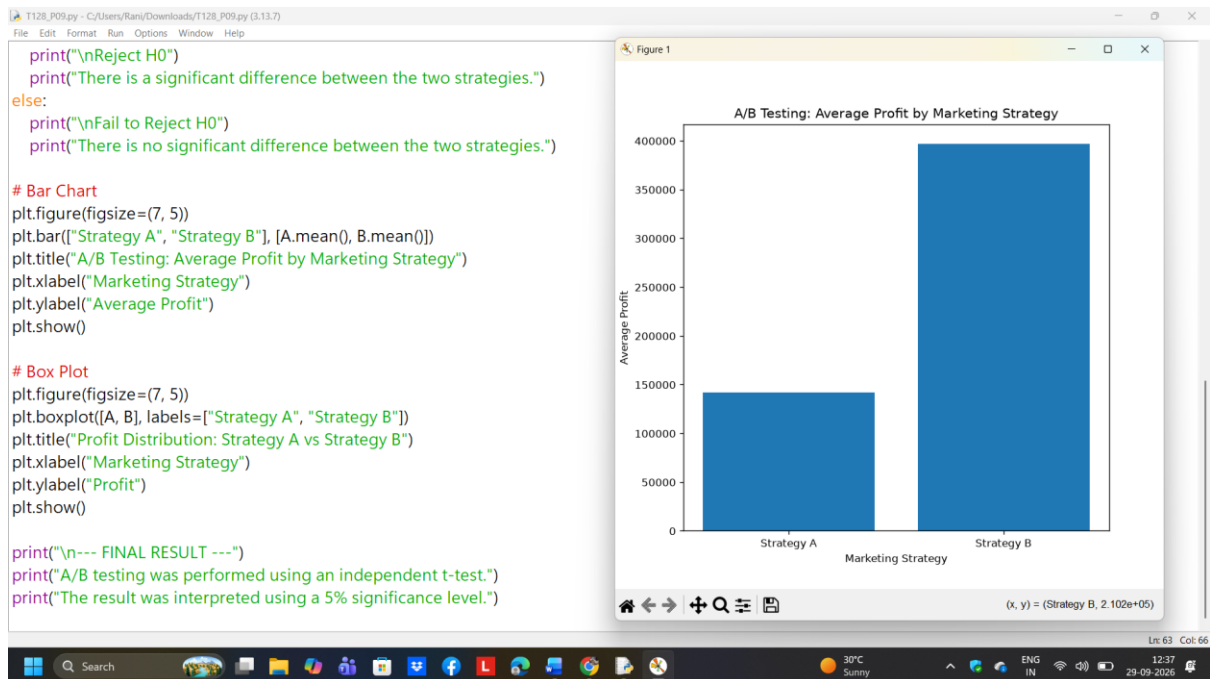
===== RESTART: C:/Users/Rani/Downloads/T128_P09
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y =====
Name: Rani Chavre
Roll No.: T128
AIM: Hypothesis Testing

--- DATASET ---
ID Region Marketing_Strategy Profit
0 1 Mumbai Strategy A 120000
1 2 Pune Strategy A 145000
2 3 Delhi Strategy A 135000
3 4 Nashik Strategy A 155000
4 5 Thane Strategy A 128000

--- HYPOTHESIS ---
H0: There is no significant difference between Strategy A and Strategy
H1: There is a significant difference between Strategy A and Strategy B.

--- AVERAGE PROFIT ---
Strategy A Average Profit: 141866.67
Strategy B Average Profit: 397000.0

--- A/B TEST RESULT ---
t-statistic: -21.8295
p-value: 0.0
```



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